



AI-Powered Smart Waste Management Platform

Revolutionizing waste sorting efficiency with artificial intelligence and data analytics

Developed by: **Achraf Oudamane**

 Next.js 14 • TypeScript • TailwindCSS •  Gemini AI

The Global Waste Crisis

Critical challenges affecting our environment and sustainability efforts worldwide:



Low Public Awareness

Insufficient knowledge about proper waste disposal



Inefficient Sorting

Poor waste sorting practices leading to contamination



Suboptimal Collection

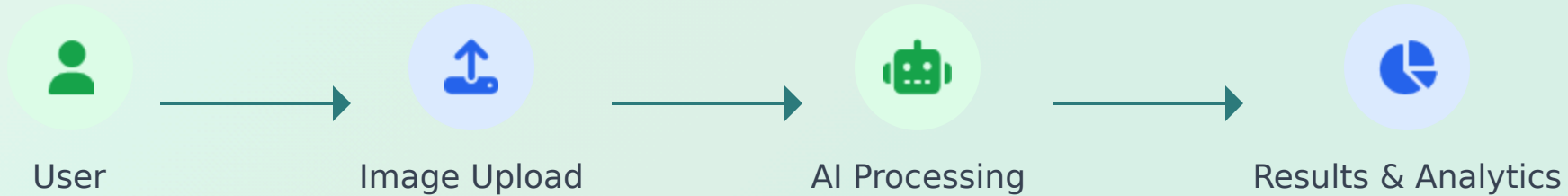
Inefficient waste collection processes

Impact on Environmental Sustainability



Our AI-Driven Solution

Intelligent platform combining AI image recognition with dashboard analytics to revolutionize waste identification and management processes.



Accurate Identification

AI-powered classification with high accuracy rates for proper waste categorization



Data-Driven Insights

Comprehensive analytics dashboard for tracking waste patterns and improving collection



Environmental Impact

Promoting eco-friendly behaviors and resource conservation through awareness

Key Platform Features

A comprehensive set of tools designed to revolutionize waste management through AI technology



Waste Photo Upload

Easily upload images of waste items for instant classification and analysis



AI Classification

Advanced Gemini AI technology accurately identifies waste types and provides detailed insights



Detailed Results

Clear categorization and information about proper disposal methods



Interactive Dashboard

Visualize waste patterns and trends with comprehensive analytics tools



Responsive Design

Seamless experience across all devices with clean, modern interface

Features continuously updated based on user feedback and emerging technologies

Technology Stack

Modern development stack for robust performance and seamless integration:



Frontend

User interface with Next.js 14 and TailwindCSS



AI Integration

Gemini AI Vision API for waste classification



Language

TypeScript for type-safe and maintainable code

System Workflow

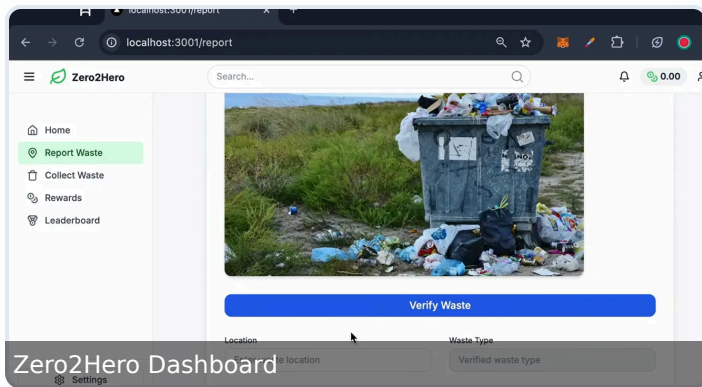
Complete user journey from image upload through AI processing to classification results and dashboard insights generation.



Platform Demo

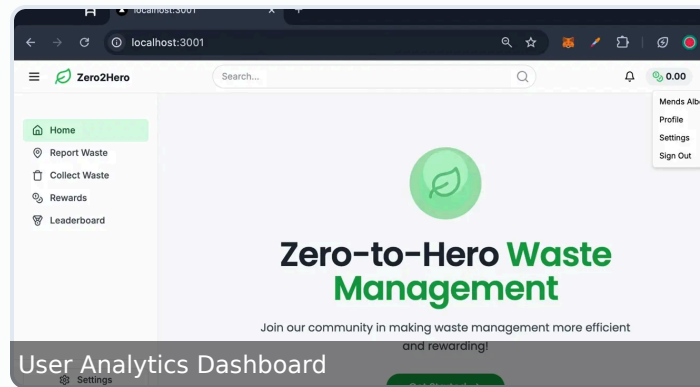
Experience the Zero2Hero platform interface and key functionality:

MAIN INTERFACE



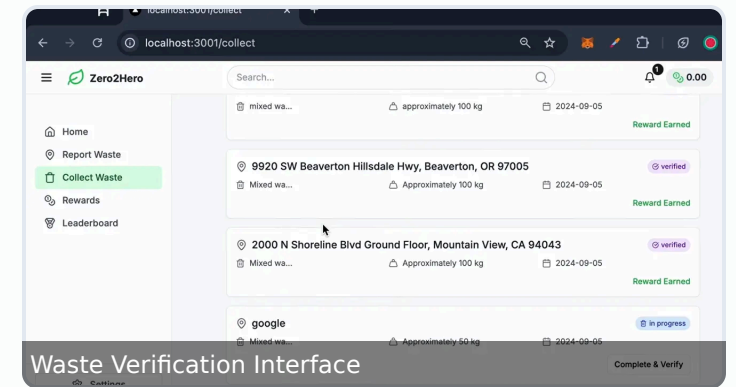
- ✓ Intuitive navigation menu
- ✓ Waste reporting functionality
- ✓ Community leaderboard

USER DASHBOARD



- ✓ Personal waste tracking
- ✓ Classification history
- ✓ Eco-points accumulation

WASTE VERIFICATION



- ✓ Image-based waste identification
- ✓ Location verification
- ✓ Weight estimation

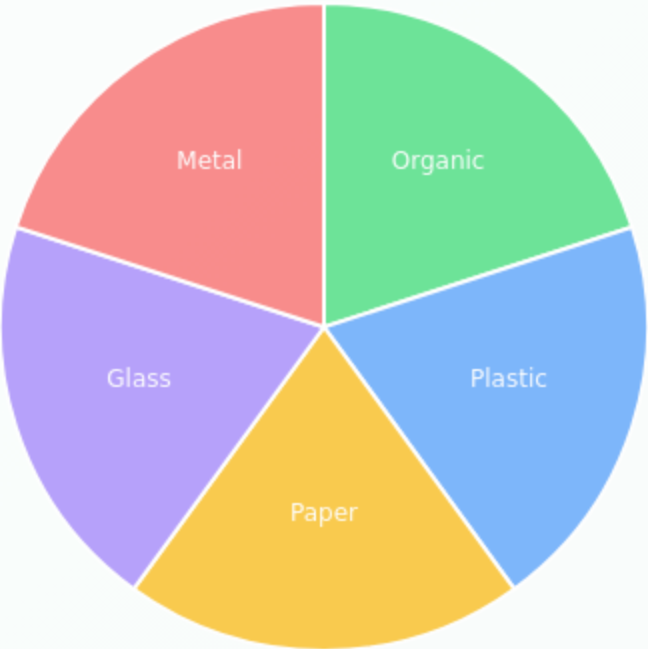
The platform is built with Next.js 14, TypeScript, and TailwindCSS

Dashboard Analytics

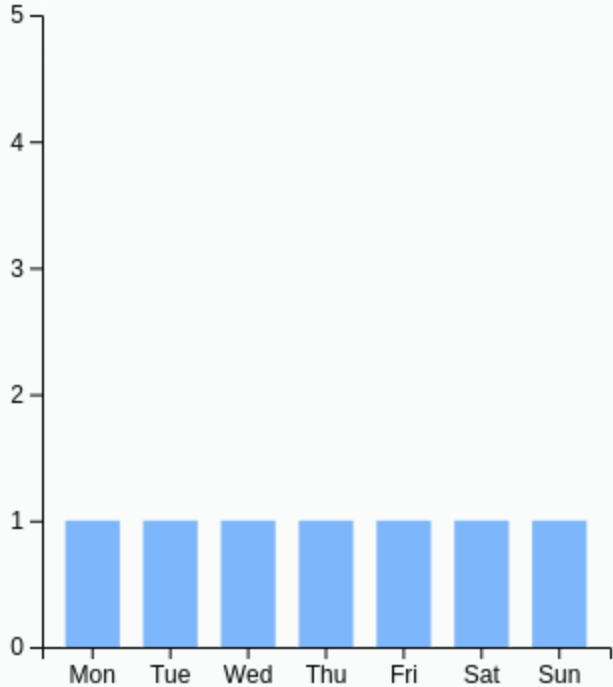
Key metrics for waste management optimization:



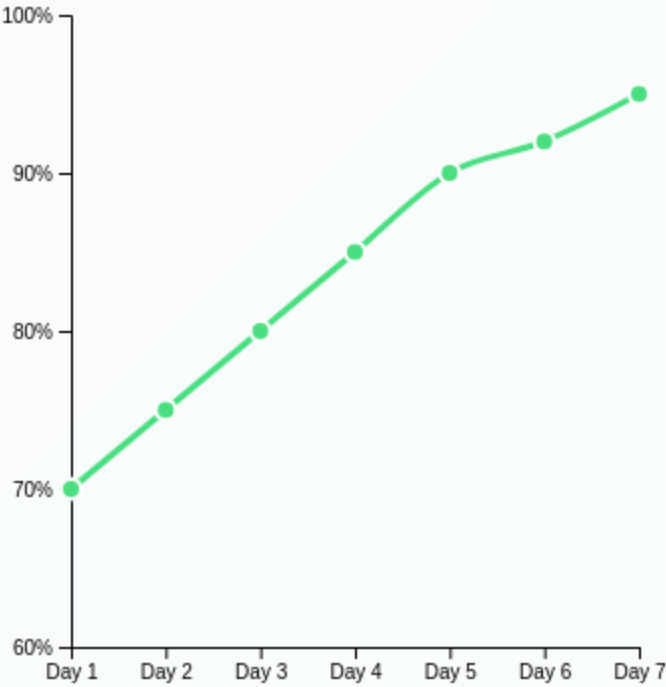
Waste Distribution



Daily Uploads



Classification Accuracy



Data updates in real-time as new waste classifications are processed

Development Challenges

Technical obstacles overcome during development:



AI API Integration

Integrating Gemini AI for waste classification.



Image Processing

Handling various image types and sizes.



Dashboard Implementation

Creating intuitive data visualizations.



Application Architecture

Designing a clean routing structure.

Overcoming Challenges



Impact & Future Vision

Current Impact

Our AI-Powered Smart Waste Management Platform aims to modernize waste sorting efficiency and promote eco-friendly behaviors through intuitive AI-driven classification and actionable analytics.

Future Enhancements



Mobile Application

Develop native mobile apps for iOS and Android to enable on-the-go waste identification and reporting.



Rewards System

Implement a gamified points system to incentivize eco-friendly waste disposal behaviors.



Smart Bin Mapping

Create a network of smart waste bins with location tracking and fill-level monitoring.

Thank you for your interest!